BABEŞ BOLYAI UNIVERSITY, CLUJ NAPOCA, ROMÂNIA
FACULTY OF MATHEMATICS AND COMPUTER SCIENCE

AI-Enabled Cardiac Digital Twin from Wearable Sensors

Personalised, multi-agent monitoring of single-lead ECG

– Advanced Machine Learning report –

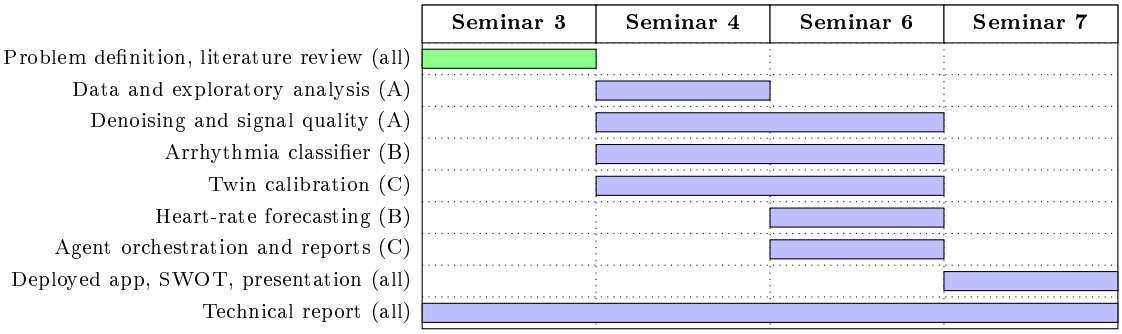
Milestone 1 (Seminar 3): problem description and survey of related work

Authors
[Member A], [specialisation], [group], [email]
[Member B], [specialisation], [group], [email]
[Member C], [specialisation], [group], [email]

2026-2027

Team

Member	Role	Milestone 1 tasks
[Member A], [email], [specialisation]	Signal Monitoring agent, data pipeline	Survey: wearable ECG, denoising
[Member B], [email], [specialisation]	Arrhythmia Detection and Physiological Prediction agents	Survey: classification, forecasting
[Member C], [email], [specialisation]	Twin Calibration and Clinical Recommendation agents	Problem definition; survey: twins, agents



Work plan (A, B, C: responsible members; green: done).

Abstract

Wearables record a single-lead ECG continuously, but the signal is noisy and differs from person to person. We propose a lightweight cardiac digital twin: a 15-parameter dynamical ECG model calibrated to the wearer's own beats and shared by five agents for signal monitoring, arrhythmia detection, rhythm forecasting, twin calibration and clinical recommendation. The twin's deviation score flags beats that are unusual for the patient. This report defines the problem formally (graphical abstract in Figure 2.1) and surveys related work. Experiments will use the MIT-BIH Arrhythmia and Noise Stress Test databases with an inter-patient split.

Contents

1 Introduction 1

2 Scientific Problem 1

3 State of the art/Related work 4

4 Conclusion and future work 7

1 Introduction

1.1 What? Why? How?

Cardiovascular diseases are the leading cause of death worldwide, with an estimated 19.8 million deaths in 2022 [36]. Smartwatches can already screen large populations for atrial fibrillation [26], but single-lead wearable ECG is noisy and of limited diagnostic value on its own [2]. Digital twins, patient-specific models updated from the patient's own data, are a promising way to personalise monitoring [7]. Current cardiovascular twins, however, are mostly simulation models built from clinical data [6, 10].

We build a lightweight twin from wearable data alone: a dynamical ECG model [20] calibrated to the wearer's beats, combined with learned models for denoising, arrhythmia detection and heart-rate forecasting. Five agents share the twin and write a report for a clinician. The key addition is a per-beat score of how far each beat is from the patient's own normal.

1.2 Paper structure and original contribution(s)

Chapter 2 defines the problem, the five agent sub-problems and the evaluation protocol. Chapter 3 surveys related work, and Chapter 4 lists the next steps.

2 Scientific Problem

2.1 Problem definition

Wearable ECG is hard to use for three reasons:

- **Noise.** Motion, muscle and electrode artefacts corrupt the signal, so its quality has to be checked before any analysis [29].
- **Volume.** A single lead at 250 Hz gives 21.6 million samples per day.
- **Inter-patient variability.** Classifiers tested on unseen patients do much worse, especially on supraventricular beats [8], which led to patient-specific models [15].

A digital twin is a model of one individual that is continually updated from their data and used for monitoring and decision support [7, 25]. Existing cardiac twins are biophysical models built from imaging and 12-lead ECG [10], data a wearable does not provide. We instead use a low-dimensional

ECG model [20] calibrated to the wearer's clean beats, together with learned models, split across the five agents of the course brief. The output is a report for a clinician; the system does not diagnose.

Why an intelligent algorithm? Fixed filters and thresholds break down under noise and individual variation. On single-lead ambulatory ECG, a deep network exceeds the average cardiologist's F_1 score [12]. Learning is also needed for denoising, for forecasting heart rate and for calibrating the twin, which is an inverse problem. The agent layer decides which models to run and when the signal is too poor to trust.

2.2 Formal problem statement

Signals. For patient p the wearable records $\tilde{x}_p[n] \in \mathbb{R}^C$ (usually $C = 1$) at f_s Hz, possibly with auxiliary signals $s_p[n]$ (optical heart rate, motion). We assume

$$\tilde{x}_p[n] = x_p[n] + \eta[n], \tag{2.1}$$

where x_p is the clean ECG and η is baseline wander, muscle and electrode-motion artefact (the noise types of the MIT-BIH Noise Stress Test Database [23]) plus powerline interference. R peaks r_k give RR intervals $RR_k = r_k - r_{k-1}$ and beats $b_k \in \mathbb{R}^{L \times C}$.

The twin. The twin of patient p at time t is

$$\mathcal{T}_p(t) = \big(\vartheta_p(t),\ \rho_p(t),\ \mathcal{H}_p(t)\big), \tag{2.2}$$

with ECG model parameters ϑ_p, rhythm state ρ_p (mean heart rate, variability) and history of findings $\mathcal{H}_p$. In the McSharry model [20] a point circles a unit limit cycle in the (x, y) plane with angular velocity $\omega = 2\pi/RR$, and the ECG is its z coordinate:

$$\dot{z} = -\sum_{i \in \{P,Q,R,S,T\}} a_i\, \Delta\theta_i \exp\!\left(-\frac{\Delta\theta_i^2}{2b_i^2}\right) - \big(z - z_0(t)\big), \qquad \Delta\theta_i = (\theta - \theta_i) \bmod 2\pi. \tag{2.3}$$

Each wave is a Gaussian with angle θ_i, amplitude a_i and width b_i, so $\vartheta \in \mathbb{R}^{15}$; z_0 is a respiratory baseline. $g(\vartheta, RR)$ denotes one generated beat.

Sub-problems. Each agent solves one sub-problem (Table 2.1). Calibration (P4) estimates the patient's parameters from clean beats classified as normal, $\mathcal{K}_p$, with a prior around the population

parameters ϑ_0:

$$\hat{\vartheta}_p = \arg\min_{\vartheta} \sum_{k \in \mathcal{K}_p} \left\| b_k - g(\vartheta, RR_k) \right\|_2^2 + \lambda \left\| \vartheta - \vartheta_0 \right\|_{\Sigma^{-1}}^2. \tag{2.4}$$

Each new beat is compared with the beat the twin predicts,

$$d_k = \frac{\left\| b_k - g(\hat{\vartheta}_p, RR_k) \right\|_2}{\hat{\sigma}_p \sqrt{L}}, \tag{2.5}$$

where $\hat{\sigma}_p$ is the patient's noise level, so $d_k \approx 1$ for typical beats. d_k is an extra input to the arrhythmia detector.

Table 2.1: Sub-problems, one per agent.

Agent	**Input → output**	**Learning problem**	**Evaluation**
(P1) Signal Monitoring	$\tilde{x} \to$ clean $\hat{x}$, quality $q \in [0,1]$	Denoiser D_ϕ, $\min_\phi \mathbb{E}\|D_\phi(\tilde{x}) - x\|^2$, trained on clean records plus recorded noise	SNR improvement, RMSE, PRD
(P2) Arrhythmia Detection	b_k, RR context, $d_k \to$ $\hat{y}_k \in \{N, S, V, F, Q\}$	Beat classification into the AAMI EC57 classes [1]	Per-class sensitivity and positive predictivity, macro-F_1
(P3) Physiological Prediction	RR history, $s \to$ quantiles of the next H RR intervals	Probabilistic multi-horizon forecasting [18]	MAE, quantile loss; vs. persistence and autoregression
(P4) Twin Calibration	clean normal beats $\to \hat{\vartheta}_p$, d_k	Eqs. (2.4)–(2.5)	Held-out RMSE; AUROC of d_k, ectopic vs. normal
(P5) Clinical Recommendation	agent outputs → report (findings, evidence, priority)	Calls the other agents as tools; abstains when q is low	Correctness of findings, latency

Evaluation. We use the inter-patient split of de Chazal et al. [8]: the 44 non-paced MIT-BIH records form two halves of 22 records, one for training and one for testing. Records 201 and 202 come from the same subject and stay together [22]. Noise robustness is tested by adding recorded noise [23] at several SNRs. Our hypothesis is that adding d_k improves the detection of S and V beats at equal positive predictivity.

2.3 Graphical abstract

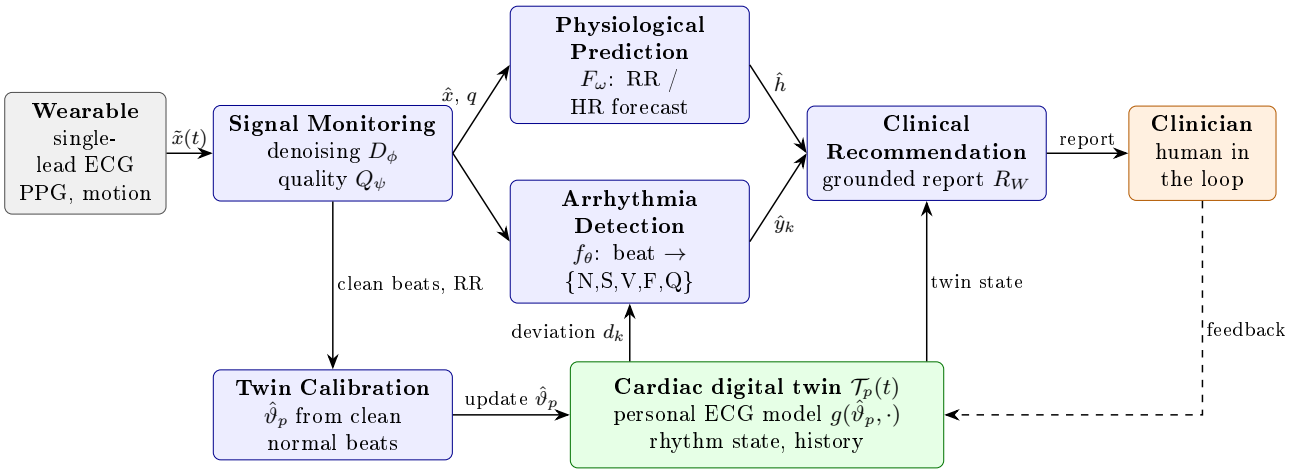


Figure 2.1: Graphical abstract: five agents share a patient-specific cardiac twin; the clinician's feedback updates the twin.

3 State of the art/Related work

We searched Crossref, Semantic Scholar, OpenAlex, PubMed and the web for work in the five areas below, covering seminal papers and recent (2022–2026) ones.

3.1 Cardiac digital twins

Corral-Acero et al. [7] define the healthcare digital twin as a tool that integrates an individual's data over time using mechanistic and statistical models, and envision twins fed by wearable data. Niederer et al. [25] note that current twins are bespoke and hard to scale. Coorey et al. [6] require bidirectional data exchange with the patient; their review of 88 papers found that existing cardiovascular "twins" were simulation models, not real-time systems.

Most implemented twins are biophysical. Gillette et al. [10] fit a patient's anatomy to MR images and electrophysiology parameters to a 12-lead ECG; functional twinning was shown for a single subject. Simonian et al. [31] build an image-based mitral-valve twin to predict the outcome of valve repair. Thangaraj et al. [35] describe hybrid twins that combine physiological parameters with lightweight deep learning, and point to smartwatches as a way to close the loop with the patient. Roop et al. [27] personalise a hybrid-automata model of the conduction system from wearable signals, and the "Cardio Twin" [19] is an edge CNN that detects ischaemic heart disease from ECG (85.77% accuracy).

Difference. Our twin is a 15-parameter ECG model calibrated from one wearable lead, much cheaper than a biophysical twin. Unlike the Cardio Twin, it is a personalised model rather than a classifier, and clinician feedback flows back into it.

3.2 Wearable ECG, signal quality and denoising

Bayoumy et al. [2] find single-lead ECG useful for common arrhythmias such as AF but not for complex arrhythmias or infarction. In the Apple Heart Study [26], 0.52% of 419,297 participants received an irregular-pulse notification; 34% of those who returned an ECG patch had AF, and patches were applied on average 13 days after the notification. Bhavsar et al. [3] treat wearable measurements as digital biomarkers and add them to clinical reports. Del-Valle-Soto et al. [9] survey biometric uses of wearables and stress privacy and data security.

Satija et al. [29] review ECG quality assessment and call for lightweight real-time methods for wearables. Smital et al. [32] grade quality continuously into three levels (full waveform, QRS only, unusable). Denoisers are usually tested on the MIT-BIH Noise Stress Test Database, where electrode-motion noise is hardest because it can mimic ectopic beats [23]. Sameni et al. [28] use a dynamical ECG model inside Kalman filters, beating band-pass, adaptive and wavelet filtering. Learned alternatives include a convolutional denoising autoencoder [4] and the diffusion model DeScoD-ECG [16].

Difference. We combine a learned denoiser with graded quality levels [32] and evaluate both by their effect on arrhythmia detection, not in isolation.

3.3 Arrhythmia classification

The MIT-BIH Arrhythmia Database has 48 two-channel half-hour recordings from 47 subjects at 360 Hz [22], and the AAMI EC57 standard defines how detectors are tested, with beat classes N, S, V, F and Q [1]. De Chazal et al. [8] introduced the inter-patient split; their feature-based classifier reached only 38.5% positive predictivity on supraventricular beats. Kiranyaz et al. [15] train a small 1-D CNN per patient on common data plus the patient's own beats.

Hannun et al. [12] classify 12 rhythm classes from single-lead ambulatory ECG with a mean ROC AUC of 0.97. Strodthoff et al. [34] benchmark models on PTB-XL, where ResNet- and Inception-style CNNs perform best. ECGFounder [17], trained on about 10.8 million ECGs, has been adapted to single-lead recordings. Kapoor and Narayanan [13] show that using the same patient in training and test data inflates results.

Difference. We keep a population classifier with a strict inter-patient split. Instead of training a model per patient [15], we add the twin's deviation score d_k, which needs no labels from the new patient.

3.4 Physiological ECG models and forecasting

McSharry et al. [20] proposed the dynamical model behind our twin (Eq. (2.3)). Clifford et al. [5] fit it to each beat for patient-specific filtering and beat clustering, treating beats the model cannot fit as artefacts. Sayadi et al. [30] track ECG waves with a model-based Kalman filter and detect premature ventricular contractions from the filter's output (99.10% accuracy).

For forecasting, the Temporal Fusion Transformer [18] gives interpretable multi-horizon forecasts and PatchTST [24] uses patches of the series as tokens. A one-layer linear model nevertheless beat Transformer forecasters on nine datasets [38], and for wearable heart rate a linear autoregressive model beat LSTMs [33].

Difference. We keep the fitted parameters as a persistent twin state that other agents query, and treat beats the model cannot fit as candidate abnormal beats rather than artefacts. Forecasters must beat persistence and autoregressive baselines.

3.5 LLM agents for health monitoring

ReAct [37] interleaves reasoning with tool calls. MDAgents [14] assigns one model or a group of models to a medical task depending on its complexity and was best on 7 of 10 medical benchmarks. PHIA [21] analyses wearable data through code generation and retrieval, with 84% accuracy on numerical questions. Hager et al. [11] found that LLMs diagnosed 2,400 real cases significantly worse than physicians and are not ready for autonomous decisions.

Difference. Our agents orchestrate validated signal models and write reports; they do not diagnose, and a clinician reviews every report.

3.6 Positioning

Table 3.1 compares representative works. We found no work that combines single-lead wearable input, a calibrated patient-specific ECG model, learned detection and an agent layer.

Table 3.1: Comparison with related work (✓: stated in the cited work).

Work	Single-lead / wearable	Patient-specific	Physiological model	Deep learning	Arrhythmia detection	LLM agents
Gillette et al. [10]		✓	✓			
Roop et al. [27]	✓	✓	✓			
Martinez-Velazquez et al. [19]				✓		
Clifford et al. [5]	✓	✓	✓		✓	
Sayadi et al. [30]	✓		✓		✓	
Kiranyaz et al. [15]	✓	✓		✓	✓	
Hannun et al. [12]	✓			✓	✓	
Merrill et al. [21]	✓					✓
This project	✓	✓	✓	✓	✓	✓

4 Conclusion and future work

Next steps:

- **Seminar 4:** exploratory analysis of the MIT-BIH Arrhythmia and Noise Stress Test databases; first denoiser, inter-patient classifier and forecasting baselines; twin calibration on real beats.
- **Seminar 6:** agent orchestration and clinical reports; classifier with vs. without the deviation score d_k.
- **Seminar 7:** deployed application, SWOT analysis and presentation.

Bibliography

[1] Association for the Advancement of Medical Instrumentation (AAMI). *ANSI/AAMI EC57:2012/(R)2020; Testing and reporting performance results of cardiac rhythm and ST segment measurement algorithms.* AAMI, 2013. doi:10.2345/9781570204784.

[2] Karim Bayoumy, Mohammed Gaber, Abdallah Elshafeey, Omar Mhaimeed, Elizabeth H. Dineen, Francoise A. Marvel, Seth S. Martin, Evan D. Muse, et al. Smart wearable devices in cardiovascular care: where we are and how to move forward. *Nature Reviews Cardiology*, 18(8):581–599, 2021. doi:10.1038/s41569-021-00522-7.

[3] Karan Bhavsar, Shivam Singhal, Vivek Chandel, Avijit Samal, Sundeep Khandelwal, Nasimuddin Ahmed, and Avik Ghose. Digital Biomarkers: Using Smartwatch Data For Clinically Relevant Outcomes. In *2021 IEEE International Conference on Pervasive Computing and Communications Workshops and other Affiliated Events (PerCom Workshops)*, pages 630–635. IEEE, 2021. doi:10.1109/percomworkshops51409.2021.9431000.

[4] Hsin-Tien Chiang, Yi-Yen Hsieh, Szu-Wei Fu, Kuo-Hsuan Hung, Yu Tsao, and Shao-Yi Chien. Noise Reduction in ECG Signals Using Fully Convolutional Denoising Autoencoders. *IEEE Access*, 7:60806–60813, 2019. doi:10.1109/access.2019.2912036.

[5] G. D. Clifford, A. Shoeb, P. E. McSharry, and B. A. Janz. Model-based filtering, compression and classification of the ECG. *International Journal of Bioelectromagnetism*, 7(1):158–161, 2005. URL: https://www.ijbem.org/volume7/number1/158-161.pdf.

[6] Genevieve Coorey, Gemma A. Figtree, David F. Fletcher, Victoria J. Snelson, Stephen Thomas Vernon, David Winlaw, Stuart M. Grieve, Alistair McEwan, et al. The health digital twin to tackle cardiovascular disease—a review of an emerging interdisciplinary field. *npj Digital Medicine*, 5(1), 2022. doi:10.1038/s41746-022-00640-7.

[7] Jorge Corral-Acero, Francesca Margara, Maciej Marciniak, Cristobal Rodero, Filip Loncaric, Yingjing Feng, Andrew Gilbert, Joao F Fernandes, et al. The 'Digital Twin' to enable the vision of precision cardiology. *European Heart Journal*, 41(48):4556–4564, 2020. doi:10.1093/eurheartj/ehaa159.

[8] P. deChazal, M. O'Dwyer, and R.B. Reilly. Automatic Classification of Heartbeats Using ECG Morphology and Heartbeat Interval Features. *IEEE Transactions on Biomedical Engineering*, 51(7):1196–1206, 2004. doi:10.1109/tbme.2004.827359.

[9] Carolina Del-Valle-Soto, Ramon A. Briseño, Leonardo J. Valdivia, and Juan Arturo Nolazco-Flores. Unveiling wearables: exploring the global landscape of biometric applications and vital signs and behavioral impact. *BioData Mining*, 17(1), 2024. doi:10.1186/s13040-024-00368-y.

[10] Karli Gillette, Matthias A.F. Gsell, Anton J. Prassl, Elias Karabelas, Ursula Reiter, Gert Reiter, Thomas Grandits, Christian Payer, et al. A Framework for the generation of digital twins of cardiac electrophysiology from clinical 12-leads ECGs. *Medical Image Analysis*, 71:102080, 2021. doi:10.1016/j.media.2021.102080.

[11] Paul Hager, Friederike Jungmann, Robbie Holland, Kunal Bhagat, Inga Hubrecht, Manuel Knauer, Jakob Vielhauer, Marcus Makowski, et al. Evaluation and mitigation of the limitations of large language models in clinical decision-making. *Nature Medicine*, 30(9):2613–2622, 2024. doi:10.1038/s41591-024-03097-1.

[12] Awni Y. Hannun, Pranav Rajpurkar, Masoumeh Haghpanahi, Geoffrey H. Tison, Codie Bourn, Mintu P. Turakhia, and Andrew Y. Ng. Cardiologist-level arrhythmia detection and classification in ambulatory electrocardiograms using a deep neural network. *Nature Medicine*, 25(1):65–69, 2019. doi:10.1038/s41591-018-0268-3.

[13] Sayash Kapoor and Arvind Narayanan. Leakage and the reproducibility crisis in machine-learning-based science. *Patterns*, 4(9):100804, 2023. doi:10.1016/j.patter.2023.100804.

[14] Yubin Kim, Chanwoo Park, Hyewon Jeong, Yik Chan, Xuhai Xu, Daniel McDuff, Hyeonhoon Lee, Marzyeh Ghassemi, et al. MDAgents: An Adaptive Collaboration of LLMs for Medical Decision-Making. In *Advances in Neural Information Processing Systems 37*, pages 79410–79452. Neural Information Processing Systems Foundation, Inc. (NeurIPS), 2024. doi:10.52202/079017-2522.

[15] Serkan Kiranyaz, Turker Ince, and Moncef Gabbouj. Real-Time Patient-Specific ECG Classification by 1-D Convolutional Neural Networks. *IEEE Transactions on Biomedical Engineering*, 63(3):664–675, 2016. doi:10.1109/tbme.2015.2468589.

[16] Huayu Li, Gregory Ditzler, Janet Roveda, and Ao Li. DeScoD-ECG: Deep Score-Based Diffusion Model for ECG Baseline Wander and Noise Removal. *IEEE Journal of Biomedical and Health Informatics*, 28(9):5081–5091, 2024. doi:10.1109/jbhi.2023.3237712.

[17] Jun Li, Aaron D. Aguirre, Valdery Moura Junior, Jiarui Jin, Che Liu, Lanhai Zhong, Chenxi Sun, Gari Clifford, et al. An Electrocardiogram Foundation Model Built on over 10 Million Recordings. *NEJM AI*, 2(7), 2025. doi:10.1056/aioa2401033.

[18] Bryan Lim, Sercan Ö. Ark, Nicolas Loeff, and Tomas Pfister. Temporal Fusion Transformers for interpretable multi-horizon time series forecasting. *International Journal of Forecasting*, 37(4):1748–1764, 2021. doi:10.1016/j.ijforecast.2021.03.012.

[19] Roberto Martinez-Velazquez, Rogelio Gamez, and Abdulmotaleb El Saddik. Cardio Twin: A Digital Twin of the human heart running on the edge. In *2019 IEEE International Symposium on Medical Measurements and Applications (MeMeA)*, pages 1–6. IEEE, 2019. doi:10.1109/memea.2019.8802162.

[20] P.E. McSharry, G.D. Clifford, L. Tarassenko, and L.A. Smith. A dynamical model for generating synthetic electrocardiogram signals. *IEEE Transactions on Biomedical Engineering*, 50(3):289–294, 2003. doi:10.1109/tbme.2003.808805.

[21] Mike A. Merrill, Akshay Paruchuri, Naghmeh Rezaei, Geza Kovacs, Javier Perez, Yun Liu, Erik Schenck, Nova Hammerquist, et al. Transforming wearable data into personal health insights using large language model agents. *Nature Communications*, 17(1), 2026. doi:10.1038/s41467-025-67922-y.

[22] G.B. Moody and R.G. Mark. The impact of the MIT-BIH Arrhythmia Database. *IEEE Engineering in Medicine and Biology Magazine*, 20(3):45–50, 2001. doi:10.1109/51.932724.

[23] George B. Moody, W. E. Muldrow, and Roger G. Mark. A noise stress test for arrhythmia detectors. In *Computers in Cardiology*, volume 11, pages 381–384, 1984.

[24] Yuqi Nie, Nam H. Nguyen, Phanwadee Sinthong, and Jayant Kalagnanam. A Time Series is Worth 64 Words: Long-term Forecasting with Transformers. In *International Conference on Learning Representations (ICLR 2023)*, 2023. doi:10.48550/arxiv.2211.14730.

[25] Steven A. Niederer, Michael S. Sacks, Mark Girolami, and Karen Willcox. Scaling digital twins from the artisanal to the industrial. *Nature Computational Science*, 1(5):313–320, 2021. doi:10.1038/s43588-021-00072-5.

[26] Marco V. Perez, Kenneth W. Mahaffey, Haley Hedlin, John S. Rumsfeld, Ariadna Garcia, Todd Ferris, Vidhya Balasubramanian, Andrea M. Russo, et al. Large-Scale Assessment of a Smartwatch to Identify Atrial Fibrillation. *New England Journal of Medicine*, 381(20):1909–1917, 2019. doi:10.1056/nejmoa1901183.

[27] Partha Roop, Nathan Allen, and Shahab Kazemi. Tuning into my heart through wearables: Towards a formal cardiac digital twin. In *Proceedings of the International Symposium on Formal Methods and Models for System Design*, pages 111–121. ACM, 2025. doi:10.1145/3742875.3754694.

[28] Reza Sameni, Mohammad B. Shamsollahi, Christian Jutten, and Gari D. Clifford. A Nonlinear Bayesian Filtering Framework for ECG Denoising. *IEEE Transactions on Biomedical Engineering*, 54(12):2172–2185, 2007. doi:10.1109/tbme.2007.897817.

[29] Udit Satija, Barathram Ramkumar, and M. Sabarimalai Manikandan. A Review of Signal Processing Techniques for Electrocardiogram Signal Quality Assessment. *IEEE Reviews in Biomedical Engineering*, 11:36–52, 2018. doi:10.1109/rbme.2018.2810957.

[30] O. Sayadi, M.B. Shamsollahi, and G.D. Clifford. Robust Detection of Premature Ventricular Contractions Using a Wave-Based Bayesian Framework. *IEEE Transactions on Biomedical Engineering*, 57(2):353–362, 2010. doi:10.1109/tbme.2009.2031243.

[31] Natalie Simonian, Sneha Vakamudi, Mark Pirwitz, and Michael Sacks. Development of a Mitral Valve Digital Twin for Transcatheter Edge-to-Edge Repair. *Structural Heart*, 9:100560, 2025. doi:10.1016/j.shj.2025.100560.

[32] Lukas Smital, Clifton R. Haider, Martin Vitek, Pavel Leinveber, Pavel Jurak, Andrea Nemcova, Radovan Smisek, Lucie Marsanova, et al. Real-Time Quality Assessment of Long-Term ECG Signals Recorded by Wearables in Free-Living Conditions. *IEEE Transactions on Biomedical Engineering*, 67(10):2721–2734, 2020. doi:10.1109/tbme.2020.2969719.

[33] Alessio Staffini, Thomas Svensson, Ung-il Chung, and Akiko Kishi Svensson. Heart Rate Modeling and Prediction Using Autoregressive Models and Deep Learning. *Sensors*, 22(1):34, 2021. doi:10.3390/s22010034.

[34] Nils Strodthoff, Patrick Wagner, Tobias Schaeffter, and Wojciech Samek. Deep Learning for ECG Analysis: Benchmarks and Insights from PTB-XL. *IEEE Journal of Biomedical and Health Informatics*, 25(5):1519–1528, 2021. doi:10.1109/jbhi.2020.3022989.

[35] Phyllis M Thangaraj, Sean H Benson, Evangelos K Oikonomou, Folkert W Asselbergs, and Rohan Khera. Cardiovascular care with digital twin technology in the era of generative artificial intelligence. *European Heart Journal*, 45(45):4808–4821, 2024. doi:10.1093/eurheartj/ehae619.

[36] World Health Organization. Cardiovascular diseases (CVDs). WHO fact sheet, 31 July 2025, 2025. Accessed 28 October 2026. URL: https://www.who.int/news-room/fact-sheets/detail/cardiovascular-diseases-(cvds).

[37] Shunyu Yao, Jeffrey W. Zhao, Dian Yu, Nan Du, Izhak Shafran, Karthik Narasimhan, and Yuan Cao. ReAct: Synergizing Reasoning and Acting in Language Models. In *International Conference on Learning Representations (ICLR 2023)*, 2023. doi:10.48550/arxiv.2210.03629.

[38] Ailing Zeng, Muxi Chen, Lei Zhang, and Qiang Xu. Are Transformers Effective for Time Series Forecasting? *Proceedings of the AAAI Conference on Artificial Intelligence*, 37(9):11121–11128, 2023. doi:10.1609/aaai.v37i9.26317.